



Cross-NOAA Collaborators

ONMS Sound Monitoring Program Manager – Lindsey Peavey Reeves



ONMS Sound Regional Coordinators

- Lindsey Peavey Reeves (West Coast)
- Eden Zang (Pacific Islands Region)
- Tim Rowell (Eastern Region)



NRS Lead – Samara Haver



Technical Team at NCEI

- Emma Beretta - Analysis lead
- Quincy Cantu - Software developer
- Carrie Wall Bell - Passive Acoustic Data Archive lead

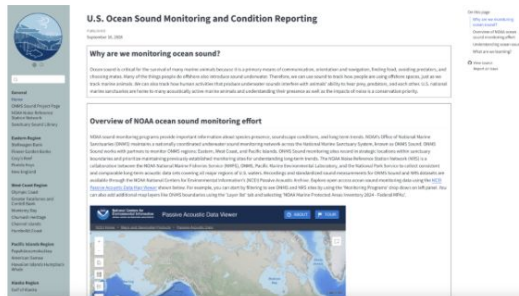


ONMS Soundscape Analyst

- Hadley Clark



Sound to Management



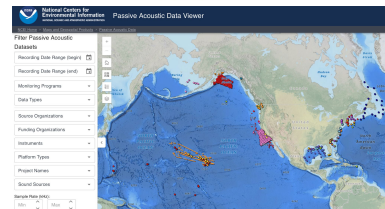
Long-term PAM Data Collection

- Passive Acoustic Monitoring (PAM)
- Ancillary data collection & access
- Opportunities to co-locate sensors
- Coordinated network



QA/QC, Standardize, Process & Archive

- Processing of high quality data
- Long-term stewardship
- Public access (NCEI)



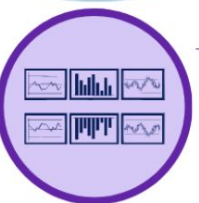
Management Applications

- Address management needs and concerns
- Decision support and interactive management tools
- Outreach, education, and stakeholder engagement
- Publications, reports, and presentations



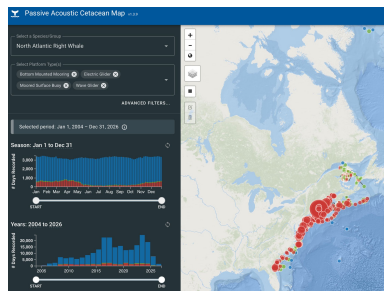
Soundscape Reporting

Long-term | Coordinated |
Standardized | Centralized |
Accessible | Supports Management



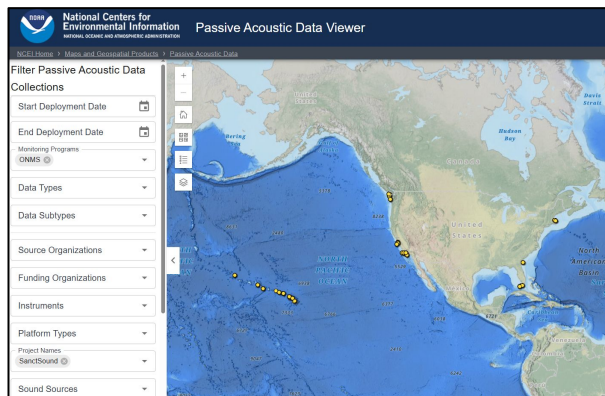
Analyze

- Comparison of long-term datasets across multiple sites
- Soundscape characterization
- Biological activity and variability
- Before-after-control-impact
- Status and trend assessment



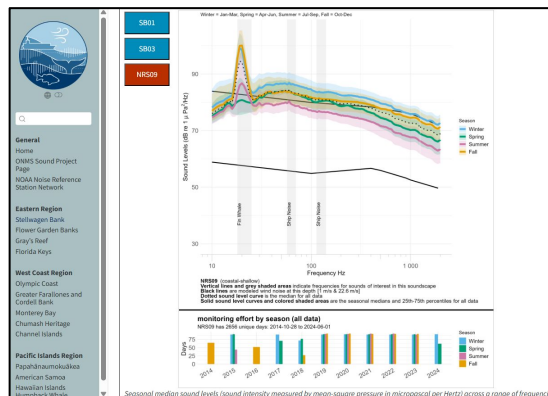
Dissemination of Data & Products: Continuum to Match Target Audience

NOAA Passive Acoustic Data Archive



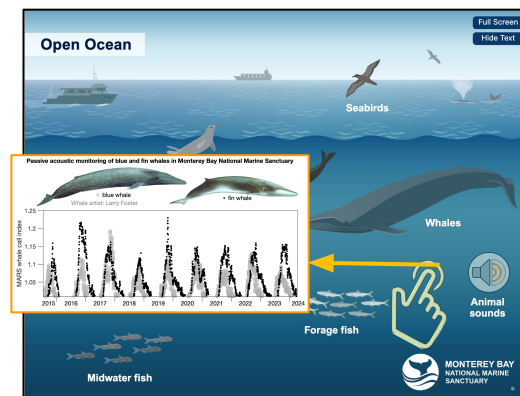
Audio data and products archived and accessible at NCEI.

U.S. Ocean Sound Monitoring Reports



Updated biannually with new data.
Enables comparison of certain indicators across MPAs and important monitoring areas.

Sanctuary Watch



Integrate site-specific information from Sound Monitoring Reports.

J. Brown, Sanctuary Watch



Level of Interpretation
Level of Technical Expertise





General

Home

ONMS Sound Project Page

NOAA Noise Reference

Station Network

Eastern Region

Stellwagen Bank

Flower Garden Banks

Gray's Reef

Florida Keys

West Coast Region

Olympic Coast

Greater Farallones and

Cordell Bank

Monterey Bay

Chumash Heritage

Channel Islands

Pacific Islands Region

Papahānaumokuākea

American Samoa

Hawaiian Islands Humpback

Whale

U.S. Ocean Sound Monitoring and Condition Reporting

PUBLISHED

August 20, 2026



SoundscapeMonitoring.us

Why are we monitoring ocean sound?

Ocean sound is critical for the survival of many marine animals because it is a primary means of communication, orientation and navigation, finding food, avoiding predators, and choosing mates. Many of the things people do offshore also introduce sound underwater. Therefore, we can use sound to track how people are using offshore spaces, just as we track marine animals. We can also track how human activities that produce underwater sounds interfere with the ability to hear prey, predators, and each other. U.S. national marine sanctuaries are home to many acoustically active marine animals and understanding the impacts of noise is a conservation priority.

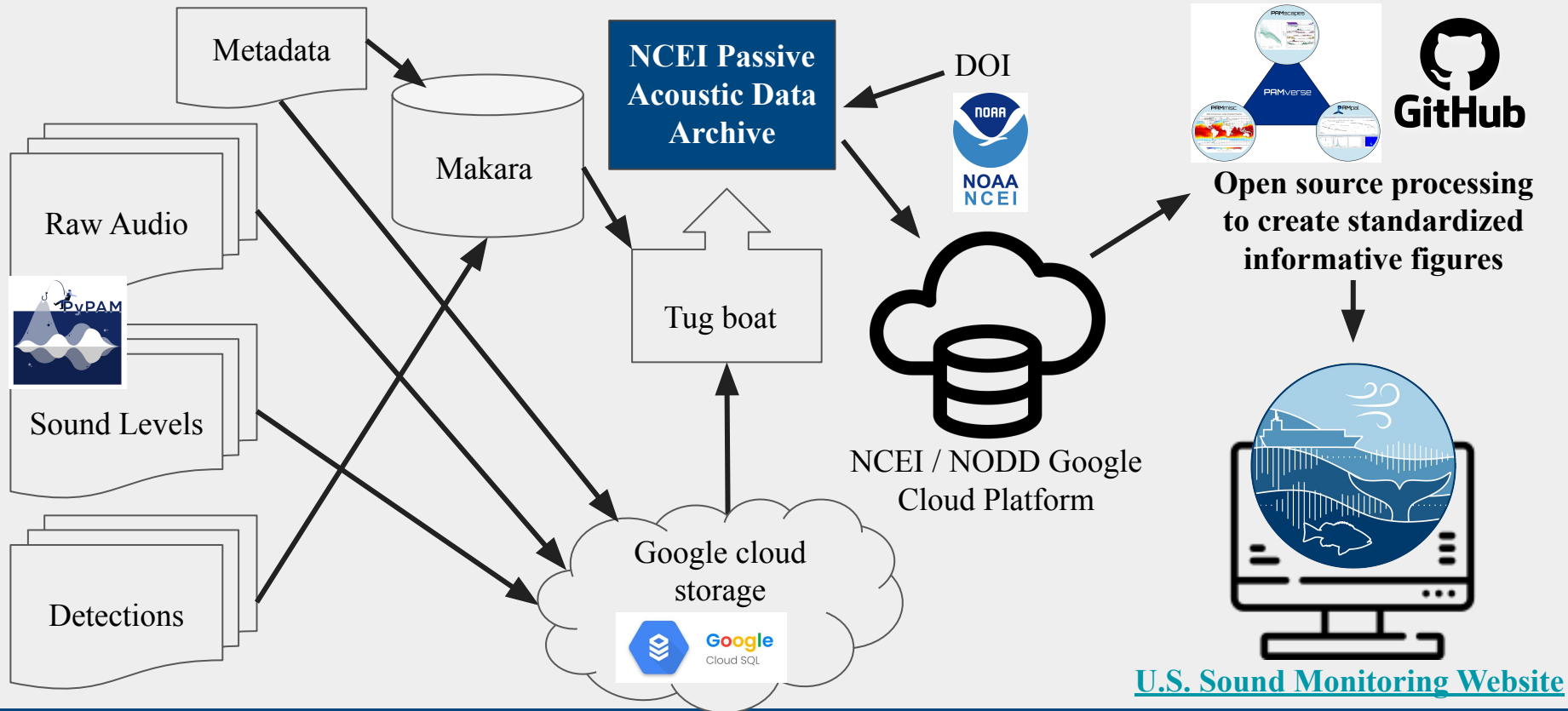
Overview of NOAA ocean sound monitoring effort

NOAA sound monitoring programs provide important information about species presence, habitat conditions, and long-term trends. NOAA's Office of National Marine Sanctuaries (ONMS) maintains a nationally coordinated underwater sound monitoring network across the National Marine Sanctuary System, known as ONMS Sound. ONMS Sound works with partners to monitor ONMS regions: Eastern, Western, and Pacific Islands. ONMS Sound monitoring sites record in strategic locations within sanctuary boundaries and prioritize maintaining previously established monitoring sites for understanding long-term trends. The NOAA Noise Reference Station Network (NRS) is a collaboration between the NOAA National Ocean Service (NMFS), ONMS, Pacific Marine Environmental Laboratory, and the National Park Service to collect consistent and comparable long-term sound data across all major regions of U.S. waters. Recordings and standardized sound measurements for ONMS Sound and NRS datasets are available through the National Centers for Environmental Information's (NCEI) Passive Acoustic Archive. Explore open access ocean sound monitoring data using the [NCEI Passive Acoustic Data Map Viewer](#) shown below. For example, you can start by filtering to see ONMS and NRS sites by using the 'Monitoring Programs' drop down on left panel. You can also add additional map layers like ONMS boundaries using the 'Layer list' tab and selecting 'NOAA Marine Protected Areas Inventory 2024 - Federal MPAs'.

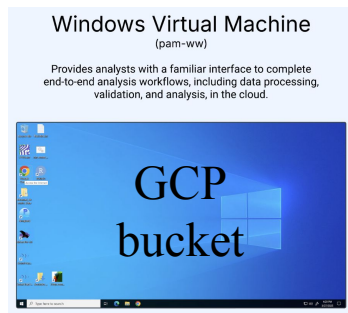




NATIONAL MARINE SANCTUARIES

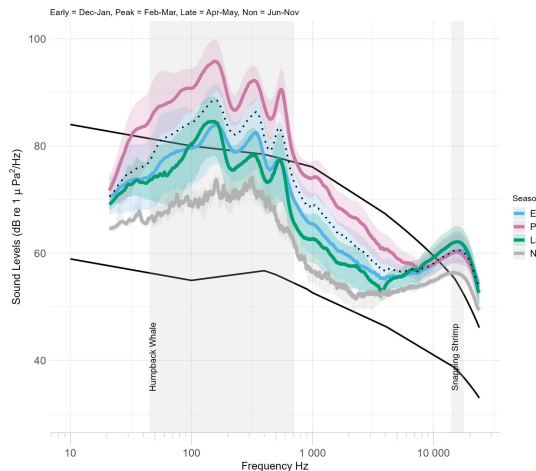


Fisheries
Google Cloud
Workstation (pam-ww)

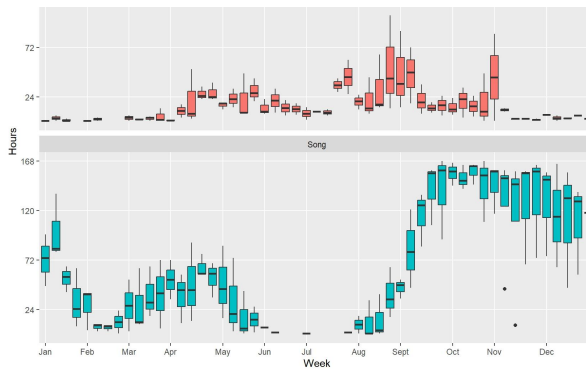


Hybrid
Millidecade
sound level
metrics

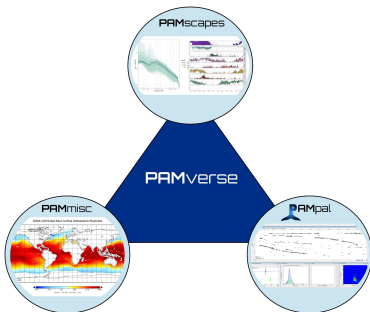
Process and create ocean sound
condition graphs in R



Create ocean sound indicator,
e.g., MBNMS humpback boxplots



PAMscapes in R

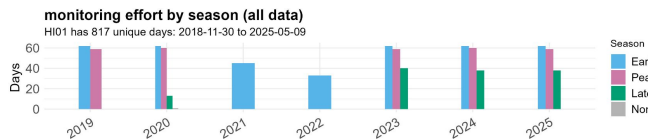
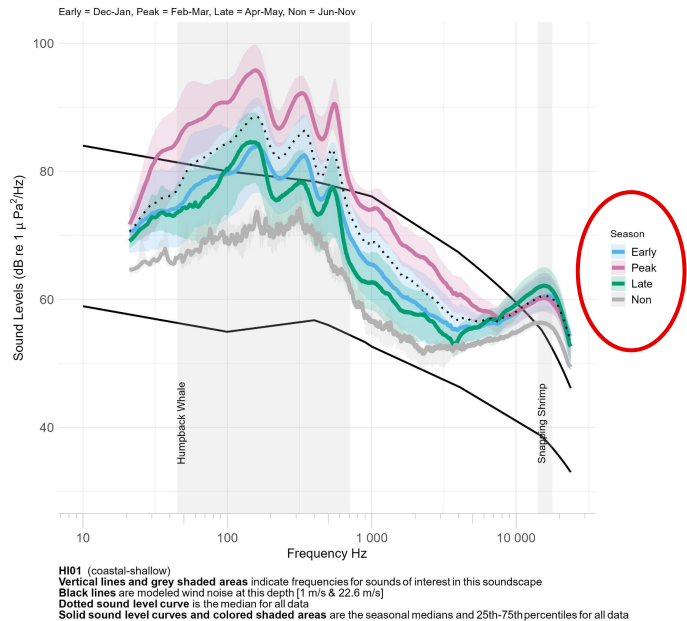


Species
Detections

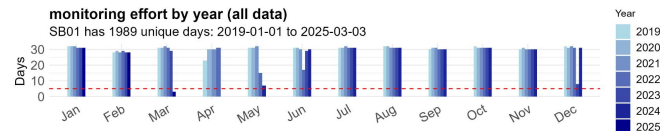
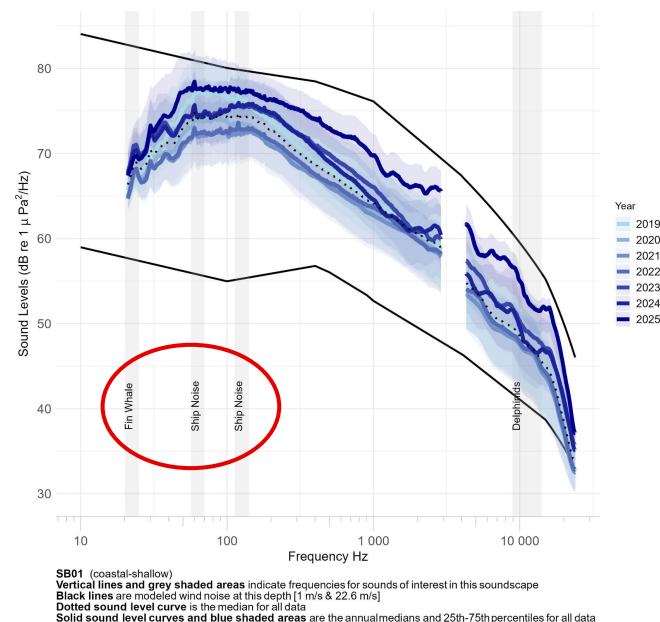
What are the seasonal patterns across frequencies?

Visualize long term trends; identify drivers of variability such as changes to:

- Species distributions
- Environmental conditions
- Anthropogenic activity
- Responses to different management actions



How are ocean sound conditions changing across years?



Stellwagen Bank National Marine Sanctuary

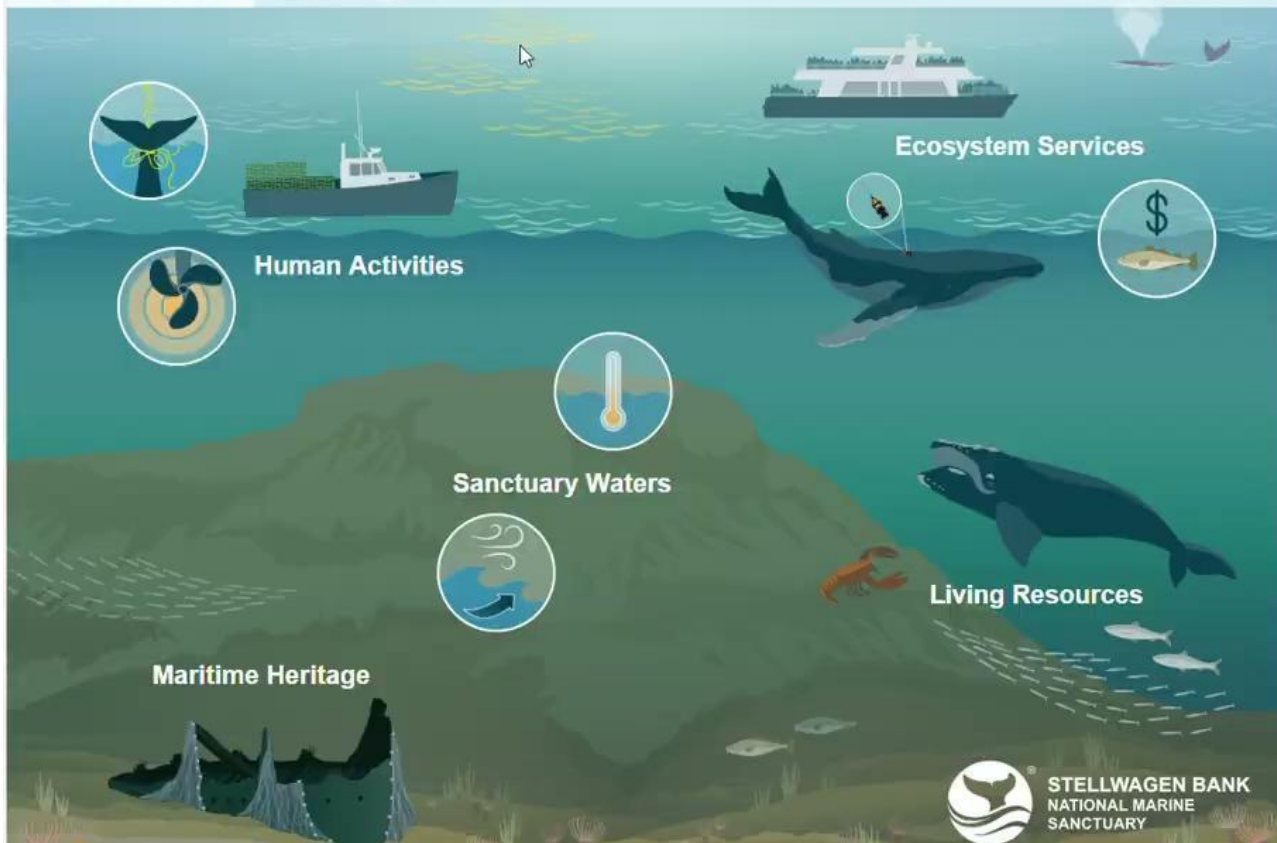
[Overview](#) [Human Activities](#) [Living Resources](#) [Sanctuary Waters](#) [Ecosystem Services](#)

Overview

Full Screen

Hide Text

- [Ecosystem Services](#)
- [Human Activities](#)
- [Living Resources](#)
- [Maritime Heritage](#)
- [Sanctuary Waters](#)



Ecosystem Services

Human Activities

Sanctuary Waters

Living Resources

Maritime Heritage



STELLWAGEN BANK
NATIONAL MARINE
SANCTUARY



General

Home

ONMS Sound Project Page

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General

Home

ONMS Sound Project Page

NOAA Noise Reference

Station Network

Sanctuary Sound Library

Eastern Region

Stellwagen Bank

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Humboldt Coast

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American Samoa

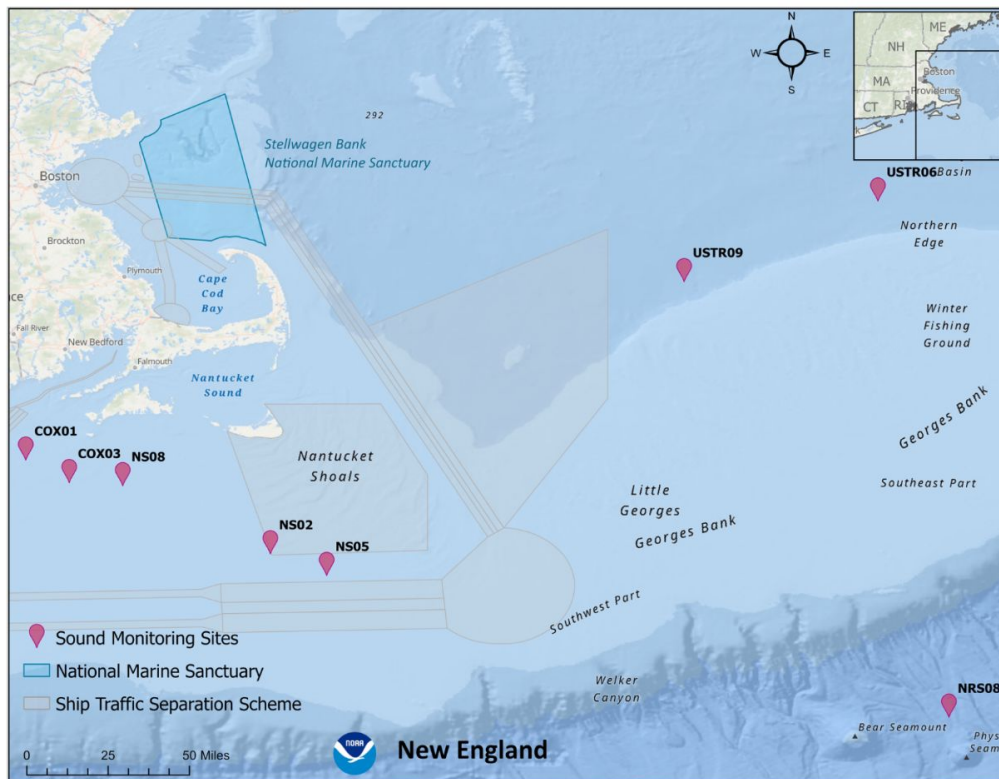
Hawaiian Islands Humpback

Whale

Alaska Region

Gulf of Alaska

Summary of monitoring sites



Long-term Monitoring Site	Primary Monitoring Purpose	Oceanographic Setting	Depth (m)	Sampling Rate (kHz)	Known Biological Sounds	Vessel Traffic Setting	Latitude (°)	Longitude (°)	Start Date	Total Recording Days	Funding Partners	Short name
NRS08	To monitor long-term trends in ocean noise, particularly from vessels and baleen whale species presence.	offshore-deep	900	5	baleen whales	commercial shipping traffic moving to and from mid-Atlantic and New England ports	40.0	-67.0	2014-06-11	1657	NMFS, PMEL, ONMS	NRS08
COX01	Alertness and monitoring	offshore	21	40	baleen	commercial and recreational	41.14126	-71.102121	2020	1421	NMFS	COX01



General

Home
ONMS Sound Project Page
NOAA Noise Reference
Station Network
Sanctuary Sound Library

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Flower Garden Banks
Gray's Reef
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New England

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Monterey Bay
Chumash Heritage
Channel Islands
Humboldt Coast

New England Soundscape Report

PUBLISHED

September 18, 2026

Update To Be Published Oct 9th

Ocean Sound Monitoring

The [Northeast Fisheries Science Center \(NEFSC\) Passive Acoustics Branch \(PAB\)](#) uses passive acoustic monitoring tools to study marine mammals, fishes, and anthropogenic sounds and provide important information for conservation and animal behavior. NEFSC PAB researchers work closely with regional partners, including Stellwagen Bank National Marine Sanctuary, to sample acoustic environments throughout Northeast Atlantic coastal and offshore areas. Multiple long-term monitoring sites have been established with specific goals to evaluate presence of data deficient [marine mammal and fish species](#) and assess potential impacts of offshore [marine energy developments](#).

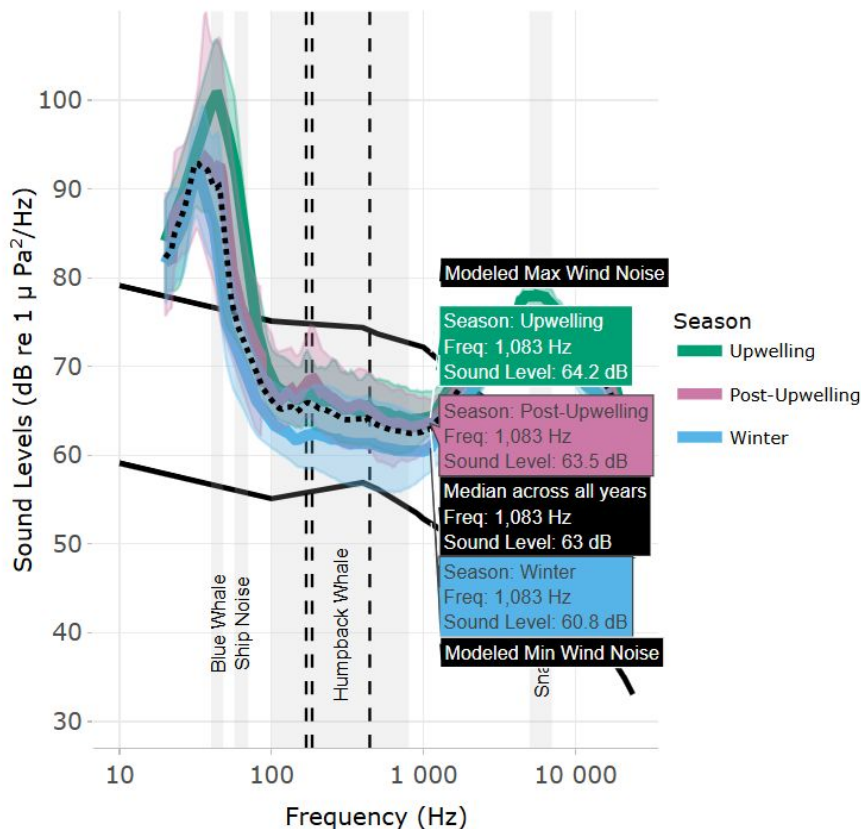
Current NEFSC ocean sound monitoring and analysis is maintained throughout the Northeast Atlantic region, and eight of those sites are featured here. An offshore site (NRS08) is part of the [U.S. Ocean Noise Reference Station Network](#) and was established in 2014 to monitor long-term trends in low-frequency ocean noise, particularly from baleen whale species presence and commercial shipping traffic moving to and from mid-Atlantic and New England ports. Two monitoring sites located on Cox Ledge (COX01, COX03) are maintained to monitor cod monitoring and to assess potential impacts of offshore wind energy development in the Southern New England area. Nearby, the monitoring stations (NS02, NS05, NRS08) expand spatial coverage for energy development monitoring and provide information about baleen whale presence, particularly North Atlantic Right Whales. To the north, two sites in the Gulf of Maine (USTR06, USTR09) have been maintained since 2022 to provide information about baleen whale presence in a previously understudied area. The Gulf of Maine is an important seasonal feeding ground for many species of endangered and protected species.

DEMO

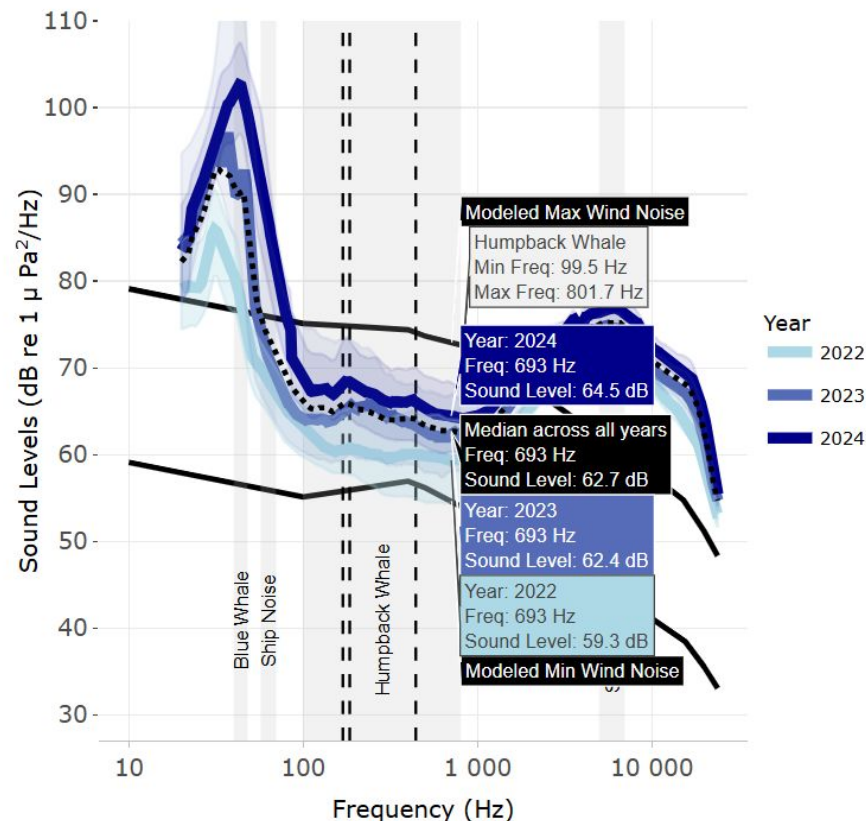
Summary of monitoring sites



Recording Site CI01, a shallow coastal environment



Recording Site CI01, a shallow coastal environment



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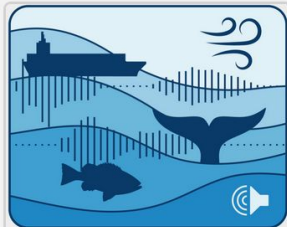
Vertical lines and grey shaded areas indicate frequencies for sounds of interest in this soundscape
Black lines are modeled wind noise at this site's depth, lower bound = 1 m/s & upper bound = 15.5 m/s
Dotted sound level curve is the median for all data
Solid sound level curves and blue shaded areas are the annual medians and 25th-75th percentiles for all data

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System-wide comparisons in development

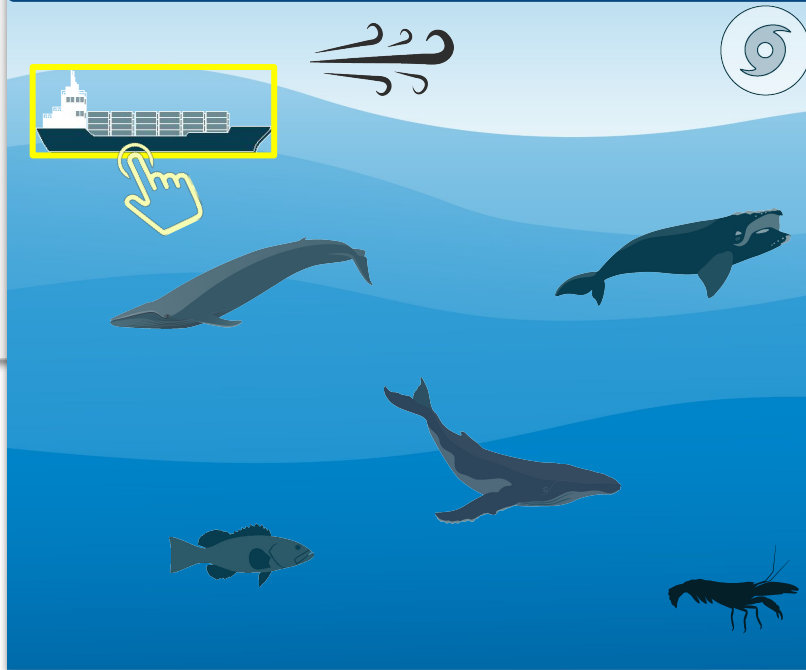
System-Wide Topics

How does the sanctuary system address topics that span multiple sanctuaries? Here we display data associated with one such topic, ocean sound.



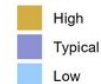
Ocean Sound

Generalized scene to access system-wide comparison of an indicator

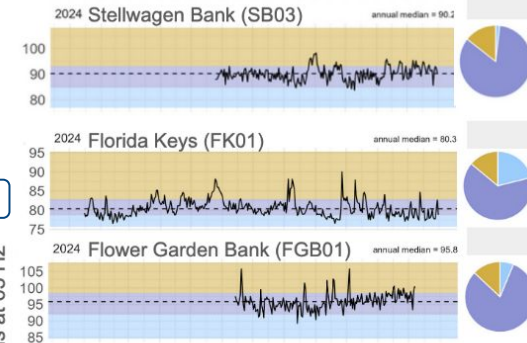


In 2024, were ship noise levels typical, higher, or lower at listening stations in these national marine sanctuaries?

Ship Noise Status



Eastern Region



West Coast Region

